

Practical assignment 1e

Assume a fair coin, and a simple pair of fair dice are being thrown.

a) Generate the probability distribution for all the possible states of the coin, and the sum of the two dice being thrown.

b) Assume that a person stands at position 0 at time 0. He throws a fair sided die. If he gets an even number, he will take that many steps forward. In other words, if he gets 2, he takes two steps forward. If he gets 4, he takes four steps forward, and if he gets 6, he takes six steps forward. On the other hand, if he gets an odd number, he takes the corresponding number of steps backwards. In other words, 1 results in one step back, 3 in 3 steps back, and 5 in 5 steps back. Generate the probability distribution for all the states after 10 dice throws.

c) After 10 dice throws, what is the probability that the person is at positions

i) 0

ii) 5

iii) 10